Where a Quantum Reservoir Works: A Transferable Operating Band

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***Abstract*—Quantum reservoir computing performs temporal machine learning by training only a linear readout on the measured outputs of a fixed quantum system, leaving the quantum dynamics itself untrained. This makes the approach attractive for near-term hardware, but it sharpens a question that has remained largely unresolved: which device settings render the fixed dynamics useful rather than merely high-dimensional. We address this question for a stateful dissipative gate-model reservoir and demonstrate that strong performance does not occupy arbitrary coordinates of the control space. Instead, it concentrates in a well-defined operating region governed by the interplay of input drive, recurrent coupling, and controlled dissipation. This region is transferable: an operating band identified on a subset of tasks predicts effective settings on held-out tasks and held-out reservoir initializations alike. Mechanism ablations establish that the region is causal rather than incidental, and an inexpensive memory-based diagnostic recovers it prior to any task-specific evaluation. The result reframes reservoir selection as the identification of a transferable, memory-preserving regime rather than the pursuit of a single benchmark optimum.**

***Index Terms*—quantum reservoir computing, dissipative dynamics, memory capacity, phase diagrams, operating regimes**

I. Introduction

Reservoir computing rests on an unusually economical idea: a fixed dynamical system, left untrained, transforms an input signal into a high-dimensional trace of its recent history, and learning reduces to fitting a single linear readout on top of that trace [1], [2], [3], [4]. The dynamics are never optimized, only observed. This economy is precisely what makes the approach appealing for near-term quantum hardware: native quantum evolution can serve as the reservoir and only the classical readout must be trained [5], [6], [7], [8]. Yet the same economy relocates the entire burden of the method onto the fixed dynamics, and so onto a question that is easy to state and surprisingly hard to answer: what makes those dynamics useful?

Recent QRC work now spans noisy gate-model devices, continuous-variable and oscillator reservoirs, weak-measurement protocols, photonic reservoirs, noise-engineered reservoirs, and experimental optical memory control [9], [10], [11], [12], [13], [14]. Classical reservoir theory has long since refused the obvious answer. Usefulness is not maximal expressivity. A reservoir earns its keep by holding a balance—retaining recent input, forgetting old input in a controlled way, and exposing nonlinear functions of that history to the readout [15], [4]. A quantum reservoir inherits this lesson but complicates it, because the three ingredients are no longer free dials. They are downstream consequences of distinct physical resources: how strongly the input perturbs the carried state, how coherently the system mixes that state across its parts, and how it is allowed to dissipate [7], [16], [17]. A large Hilbert space supplies dimension, but dimension alone preserves no memory; drive that is too strong overwrites history, and dynamics with no dissipation never forget in the way fading memory requires.

We take this seriously as an organizing principle rather than a caveat. For a stateful dissipative gate-model reservoir, we treat the space of these three resources as the proper setting in which to ask when the reservoir works, and we map it. The picture that emerges is not a single tuned operating point but a connected region—gentle input, nonmaximal coupling, controlled damping—in which good performance concentrates and from which it does not stray arbitrarily across tasks. The contribution of this paper is that region itself: its existence, its transfer across tasks and reservoir initializations, the mechanisms that sustain it, and a memory-based diagnostic that locates it before any task-specific search.

II. Protocol

Reservoir. The reservoir is a fixed quantum system that processes a scalar time series one step at a time, carrying its state forward as memory. Concretely, the state is a density matrix that is never reset between steps: each new input perturbs the state left behind by the previous step, so recent history accumulates in the system rather than being overwritten. The input enters as a small rotation whose angle is set by the current value,

$$\rho_{t,0} = U_{\rm in}(u_t)\rho_{t-1,L}U_{\rm in}^{\dagger}(u_t), \qquad U_{\rm in}(u_t) = \bigotimes_i R_y(\beta u_t). \tag{1}$$

The input strength β controls how strongly each step disturbs the carried state.

The perturbed state then evolves through a small number of identical layers. Each layer does three things in turn: it mixes each qubit locally with a frozen rotation, couples neighbors around a ring, and lets the qubits dissipate through a noise channel,

$$\begin{aligned}\rho_{t,\ell} &= \mathcal{D}_{\gamma}^{\otimes n}\left[U_{\lambda}U_{\rm mix}\rho_{t,\ell-1}U_{\rm mix}^{\dagger}U_{\lambda}^{\dagger}\right], \\ U_{\lambda} &= \prod_{(i,j)\in E_{\rm ring}} e^{-i\lambda Z_i Z_j}.\end{aligned} \tag{2}$$

The coupling strength λ sets how much stored history is mixed across qubits, and the damping rate γ sets how quickly the system forgets. These three quantities—β, λ, and γ—are the only controls we vary; every gate is otherwise frozen, and only a linear readout is trained on the measured outputs. Each control corresponds to one ingredient required by reservoir theory: β governs how new information competes with old memory, λ generates nonlinear mixing of past inputs, and γ provides the controlled forgetting on which fading memory depends. The space $\theta = (\beta, \lambda, \gamma)$ is therefore not an arbitrary grid of knobs but the natural coordinate system in which to ask whether the reservoir is useful.

Configuration. Unless stated otherwise, the reservoir uses four qubits, two layers, ring coupling, uniform input injection, amplitude damping, and a readout built from single-qubit Z and ring-pair ZZ expectation values. Training is ridge regression on these features alone. The phase grid spans $\beta/\pi = \{0, 0.02, 0.04, 0.07, 0.10, 0.22, 0.50\}$, $\lambda/\pi = \{0, 0.05, 0.10, 0.12, 0.16, 0.28, 0.42\}$, and $\gamma = \{0, 0.05, 0.12, 0.22, 0.30\}$ over reservoir seeds 42–61. Hyperparameters are selected on validation data only; holdout data never participate in defining the operating region.

Tasks, metric, and splits. We use four scalar one-step prediction tasks with chronological splits. Mackey–Glass is a chaotic delay series [18], Lorenz uses the next x coordinate of an RK4-integrated Lorenz trajectory [19], NARMA10 is a nonlinear tenth-order memory benchmark [20], and Sunspots is the annual sunspot activity series from `statsmodels`. Mackey–Glass, Lorenz, and NARMA10 use washout/train/validation/holdout lengths 300/1600/400/400; Sunspots uses 20/150/50/60. Inputs are scaled to $[-1, 1]$ and targets are standardized. The reported error is normalized mean-squared error, $\mathrm{NMSE} = \langle (y - \widehat{y})^2 \rangle / (\mathrm{Var}(y) + 10^{-12})$, computed separately on validation and holdout splits. For each reservoir setting the ridge parameter is selected only by validation NMSE from $\{10^{-8}, 10^{-7}, \ldots, 10^{3}\}$.

Operating band. To locate the useful region we rank, rather than threshold, performance. For each task j and reservoir seed s, a setting θ receives a validation percentile rank $r_{j,s}(\theta)$, with lower rank meaning better validation error. We then define the operating band as the set of settings that land in the top fraction p for at least a fraction q of all task-seed pairs,

$$B_{p,q} = \{\theta : \Pr_{j,s}[r_{j,s}(\theta) \le p] \ge q\}. \tag{3}$$

Ranks are always formed within each task-seed replicate, so task scale and seed difficulty do not dominate the aggregate. This is deliberately a frequency-level definition, weaker than demanding a single universal optimum: it asks whether a region behaves well repeatedly across tasks and initializations. In leave-one-task or leave-one-seed tests, the band is built from the remaining replicates and then scored on the withheld replicate. In the holdout audit, the validation-defined band is frozen first and then evaluated once on holdout ranks and NMSE. Confidence intervals are nonparametric bootstrap intervals for the mean with 50,000 resamples and seed 20260523.

TABLE I
HOLDOUT PERFORMANCE AFTER VALIDATION-ONLY BAND SELECTION. MEANS ARE OVER TWENTY SEEDS PER TASK; "ALL" AVERAGES THE 80 TASK-SEED ROWS.

Task	Band NMSE	Medoid NMSE	Band rank
Mackey–Glass	7.07e-05	6.08e-05	0.122
Lorenz	1.43e-05	1.80e-05	0.167
NARMA10	0.527	0.486	0.149
Sunspots	0.215	0.210	0.235
All	0.186	0.174	0.168

TABLE II
ARCHITECTURE ROBUSTNESS AFTER VALIDATION-ONLY BAND SELECTION. OVERLAP IS JACCARD OVERLAP WITH THE BASE BAND IN THE SHARED (β, λ, γ) GRID.

Setting	Band?	Held-out rank	Overlap
4q, 2 layers, Z+ZZ	yes	0.168	1.00
4q, 3 layers, Z+ZZ	yes	0.160	0.22

III. EVIDENCE

Across the control space, low validation rank does not spread arbitrarily; it gathers into a compact, connected region, and relaxing the selection threshold grows that region smoothly rather than fragmenting it—the signature of a single operating regime rather than a scatter of coincidences (Fig. 1). A region that merely fit the four tasks jointly would prove little, so we test whether it survives the removal of the information it was built from. It does: when one task is withheld and the band is defined from the rest, the held-out task still lands in the top quartile on average, with mean validation rank 0.222 and CI $[0.205, 0.239]$; when a seed is withheld instead, the held-out seed behaves the same way, with mean rank 0.162 and CI $[0.146, 0.178]$. The region is thus a search prior that generalizes across both task identity and reservoir initialization, not a fit to either.

The same validation-defined band also holds on data it never touched: after the band is frozen, evaluating once on the chronological holdout split leaves it top-quartile on unseen data (Table I). Repeating the identical protocol on a nearby four-qubit, three-layer variant tests whether the phenomenon is tied to the exact depth choice; the table reports band existence, held-out rank, and Jaccard overlap with the base control region (Table II). That the band is real rather than cosmetic is confirmed by perturbing the dynamics that sustain it: removing damping, removing recurrent mixing, or replacing amplitude damping with dephasing each erases the region entirely, with retention values 0.00, 0.00, and 0.00 under the same criterion. The collapse is total rather than gradual, exactly as expected if the regime is held in place by input strong enough to write but not overwrite, mixing that builds nonlinear features, and damping that preserves usable memory rather than merely adding noise (Fig. 2). Finally, the region can be found before any task-specific search: memory capacity correlates strongly and negatively with validation rank, $\rho_s = -0.88$, as do the information-processing capacities [15], whereas raw feature diversity does not (Fig. 3).

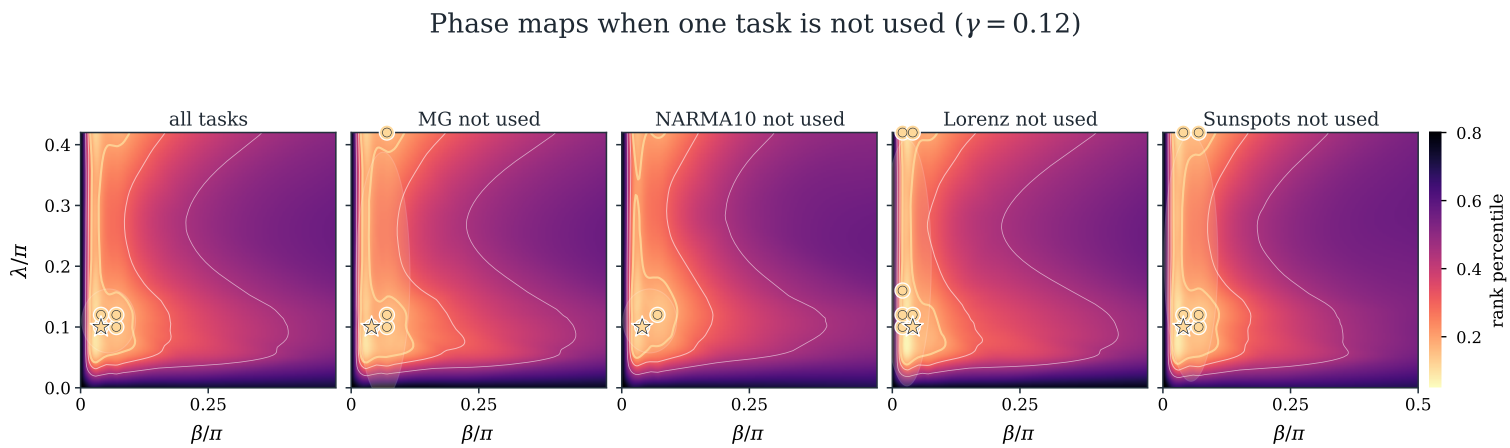

Fig. 1. Validation phase maps at the fixed $\gamma = 0.12$ slice. Bright regions are low validation percentile rank. In panels titled "X not used," task X is removed before computing the map; the map is built from the other three tasks only. Gold circles mark points that appear in the top-20% for at least 70% of the displayed task-seed replicates; the star is the medoid of the primary band.

Variant	Band pts.	Retention	Rank on base band
base AD+RxZZ	4	1.00	0.218
$\gamma = 0$	0	0.00	0.451
dephasing	0	0.00	0.453
depolarizing	0	0.00	0.511
no mixer	0	0.00	0.403
Rx only	0	0.00	0.500
ZZ only	0	0.00	0.477

IV. Discussion and Scope

For this reservoir family, useful learning is organized by a transferable operating regime rather than by task-specific tuning—a hypothesis whose credibility comes from variety. Four tasks with little in common, spanning smooth chaos, nonlinear memory, and irregular real-world data, obey the same region, and that region holds when any one task or any one seed is withheld; what survives this much variation is hard to dismiss as a fluke. The reading we draw is deflationary. The useful region is not picked out by Hilbert-space size or by feature diversity but by memory, which suggests that a quantum reservoir succeeds not because its state space is large but because drive, coupling, and dissipation together keep memory and nonlinearity in balance—the principle classical reservoir theory named long ago [15], [4]. This is also why the ablations bite: strip out damping or mixing and the regime does not weaken, it disappears. The boundaries are drawn on purpose. Everything here is simulated, on one architecture family, with measurement treated as free, so finite-shot noise, calibration drift, and readout cost are all absent and could each deform the regime [11], [21]; we therefore claim neither quantum advantage nor hardware efficiency. The claim is smaller and sturdier—that the operating regime is the right object to study, that it transfers, and that memory is how to find it.

V. Conclusion

The usefulness of a dissipative quantum reservoir is not a property of its size but of where it is operated. Across tasks and reservoir initializations, strong performance does not wander the control space; it gathers in a single memory-preserving region where input drive, recurrent coupling, and dissipation hold one another in balance. That region transfers, it survives the removal of the very mechanisms that define it only by collapsing, and it can be found in advance by a diagnostic that asks nothing about the task—only whether the reservoir remembers.

The lesson is deflationary in the most productive sense. The resource that makes this quantum reservoir work is not a quantum mystery but the same fading-memory-and-nonlinearity balance that classical reservoir theory identified long ago [15], [4], realized here through physical channels rather than chosen weights. Read this way, the contribution is methodological before it is physical: it reframes reservoir selection as the task of locating a transferable operating regime, and it offers memory, rather than raw feature diversity, as the compass that points there.

What remains is the harder half. These results are established in simulation, on a small family of architectures, with measurement treated as free; finite sampling, calibration drift, device noise, and the true cost of reading out many observables all lie outside the present claim. The natural next step is not a larger benchmark but a more honest one—to ask whether the same memory-defined regime persists once the reservoir must be measured, calibrated, and run on hardware that does not forget on command. We expect it to, and we have given the map for finding out.

Code and data availability. Code, checked-in CSV/JSON artifacts, generated figures, and the reproduction script are available in the project repository.

Reproducibility Appendix

Frozen reservoir and simulator. Exact density matrices start at $\rho_0 = |0\cdots 0\rangle\langle 0\cdots 0|$ and are carried between time steps. For seed s, `default_rng(s)` draws $\theta_i \sim U[0, 2\pi)$ once; $U_{\text{mix}} = \bigotimes_i R_x(\theta_i)$ is reused at every step and layer, with exact

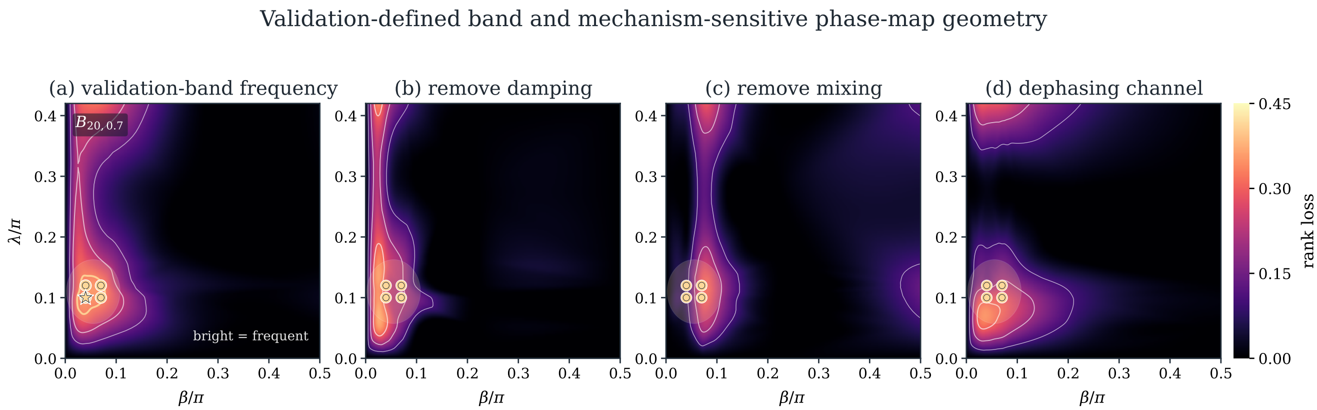

Fig. 2. QRC-only phase-map evidence for the operating-regime hypothesis. (a) The top-20 frequency map visualizes the validation-defined $B_{20,0.7}$ band at $\gamma = 0.12$: bright coordinates repeatedly rank in the top validation quintile across task-seed replicates. (b–d) Ablation rank-loss maps show how much the mean validation rank degrades when controlled damping is removed, recurrent mixing is removed, or amplitude damping is replaced by dephasing. The highlighted base-band projection lies where these mechanism removals cause rank loss.

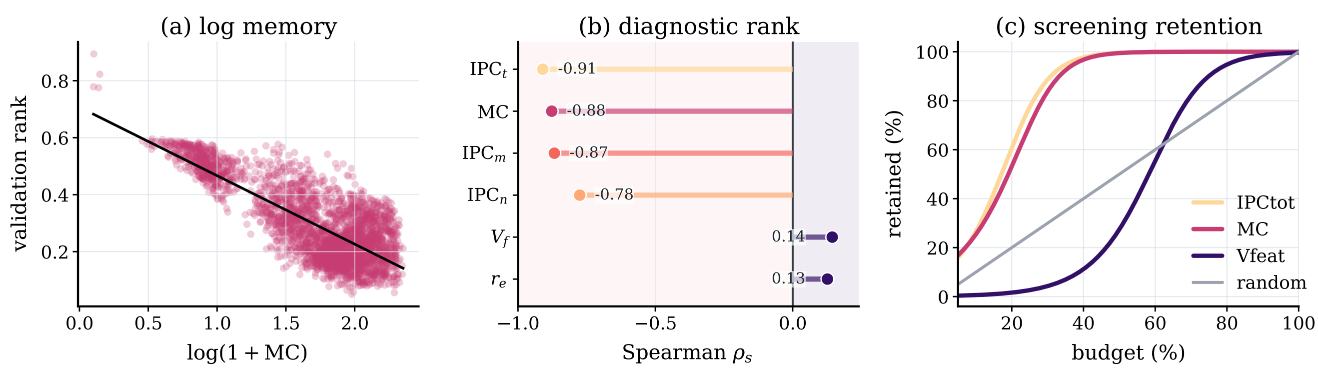

Fig. 3. Diagnostic audit for the QRC phase-map claim. A single log-memory panel expands the non-collapsed memory range and shows the strong monotone relation between retained history and validation rank. Intrinsic processing-capacity diagnostics give the strongest rank correlations, and smooth monotone guides fitted to the empirical screening table show that memory/IPC recovers useful reservoirs earlier than random or diversity-only screens.

seeds 42–61 stored in `data/frozen_rx_angles.csv`. Layers apply $U_\lambda U_{\mathrm{mix}}$ on ring edges followed by exact local amplitude damping; no finite-shot sampling is modeled. The readout is exact Z_i and ring-pair $Z_i Z_j$ expectations after two layers, with an unpenalized-intercept ridge readout.

Diagnostics and uncertainty. MC/IPC use a common iid drive $u_t \sim U[-1, 1]$ of length 900, washout 150, seed 12345, ridge $\alpha = 10^{-8}$, and a 70/30 chronological split. MC sums held-out R^2_+ over delays 1–10; IPC adds linear delays 1–20, quadratic Legendre delays 1–10, and ten cross-delay products, clipping negative R^2 to zero. Confidence intervals use 50,000 bootstrap resamples of the mean, seed 20260523. CSV/JSON data, generated macros, figures, and `./reproduce.sh` are included with the repository.

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